(e) A closed cylindrical tank is to have a surface area of 20 m^2.

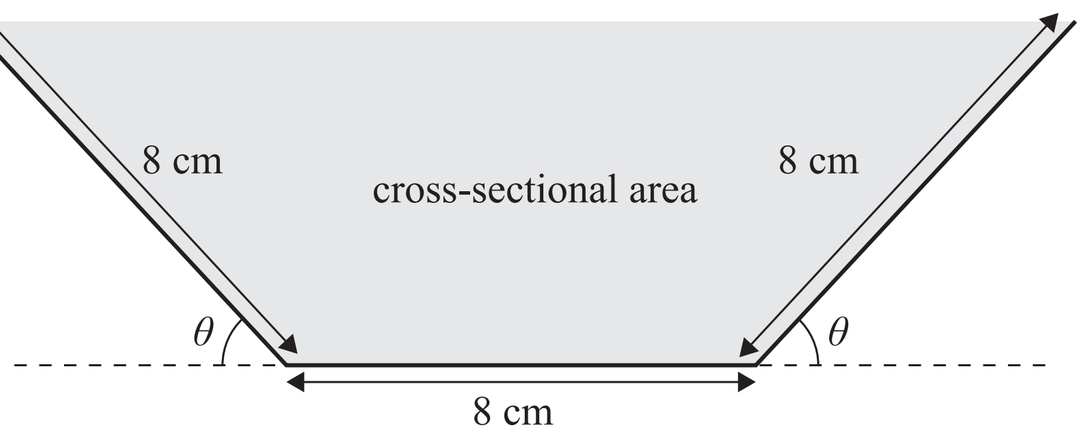
Find the radius the tank needs to have so that the volume it can hold is as large as possible.

You do not have to prove that your solution gives the maximum volume.

Show any derivatives that you need to find when solving this problem.

(e) A copper sheet of width 24 cm is folded, as shown, to make spouting.

Cross-section:



Find angle θ which gives the maximum cross-sectional area.

You do not need to prove that you have found a maximum.

Show any derivatives that you need to find when solving this problem.

(e) A spherical balloon is being inflated with helium.

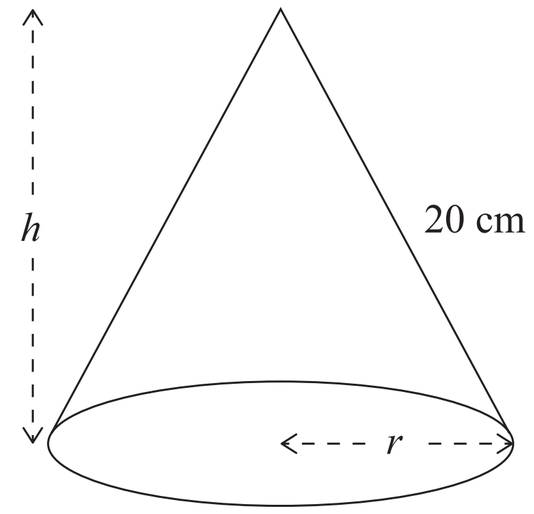
The balloon is being inflated in such a way that its volume is increasing at a constant rate of $300\ cm^3\ s^{-1}$.

The material that the balloon is made of is of limited strength, and the balloon will burst when its surface area reaches $7\,500\ cm^2$.

Find the rate at which the surface area of the balloon is increasing when it reaches bursting point.

Show any derivatives that you need to find when solving this problem.

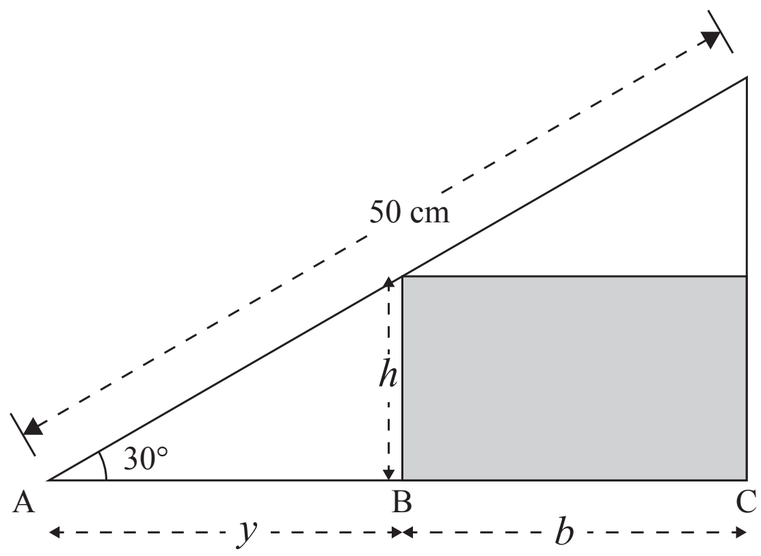
(e) What is the maximum volume of a cone if the slant length of the cone is 20 cm?



You do not need to prove that the volume you have found is a maximum.

Show any derivatives that you need to find when solving this problem.

(e) A rectangle is drawn inside a right angled triangle, as shown in the diagram below.



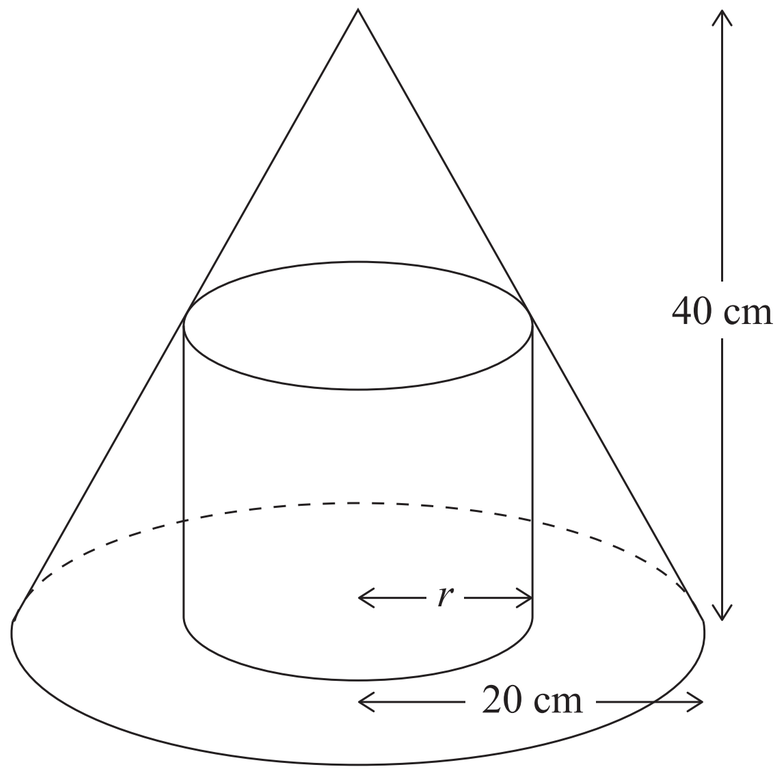
Point B moves along the base of the triangle AC, beginning at point A, at a constant speed of 3 cm s^{-1}.

At what rate is the area of the rectangle changing when point B is 20 cm from point A?

Show any derivatives that you need to find when solving this problem.

(e) A cone has a radius of 20 cm and a height of 40 cm.

A cylinder fits inside the cone, as shown below.

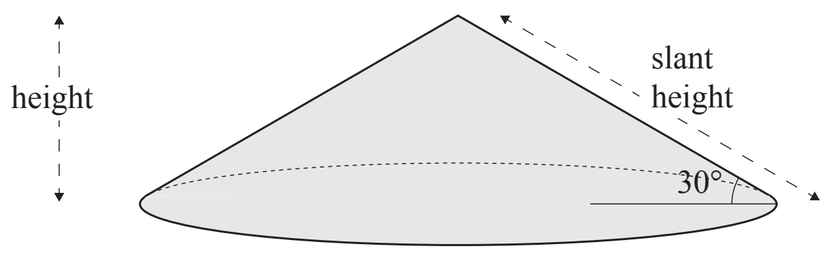


What must the radius of the cylinder be to give the cylinder the maximum volume?

You do not need to prove that the volume you have found is a maximum.

Show any derivatives that you need to find when solving this problem.

(e) Salt harvested at the Grassmere Saltworks forms a cone as it falls from a conveyor belt.
The slant of the cone forms an angle of 30° with the horizontal.
The conveyor belt delivers the salt at a rate of 2 m^3 of salt per minute.



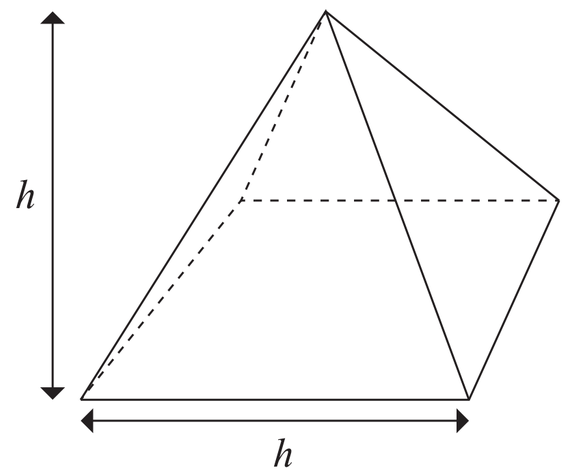
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https://bronzblog.files.wordpress.com/2013/07/imgp1182.jpg

Find the rate at which the slant height is increasing when the radius of the cone is 10 m.

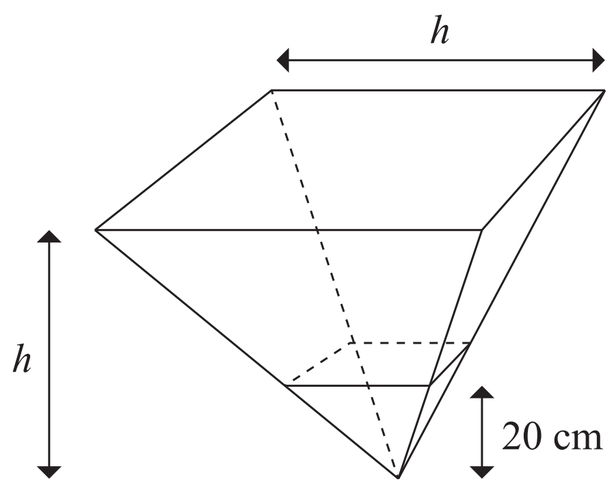
You must use calculus and show any derivatives that you need to find when solving this problem.

(e) A water container is constructed in the shape of a square-based pyramid. The height of the pyramid is the same as the length of each side of its base.



A vertical height of 20 cm is then cut off the top of the pyramid, and a new flat top added.

The pyramid is then inverted and water is poured in at a rate of 3000 cm^3 per minute.



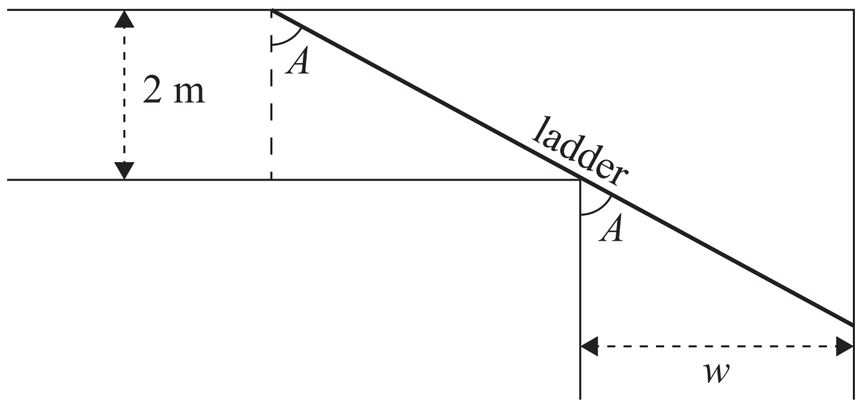
Find the rate at which the surface area of the water is increasing when the depth of the water is 15 cm.

$$\text{Volume of pyramid} = \frac{1}{3} \times \text{base area} \times \text{height}$$

You must use calculus and show any derivatives that you need to find when solving this problem.

(e) A corridor is 2 m wide.

At the end it turns 90° into another corridor.



What is the minimum width, w, of the second corridor if a ladder of length 5 m can be carried horizontally around the corner?

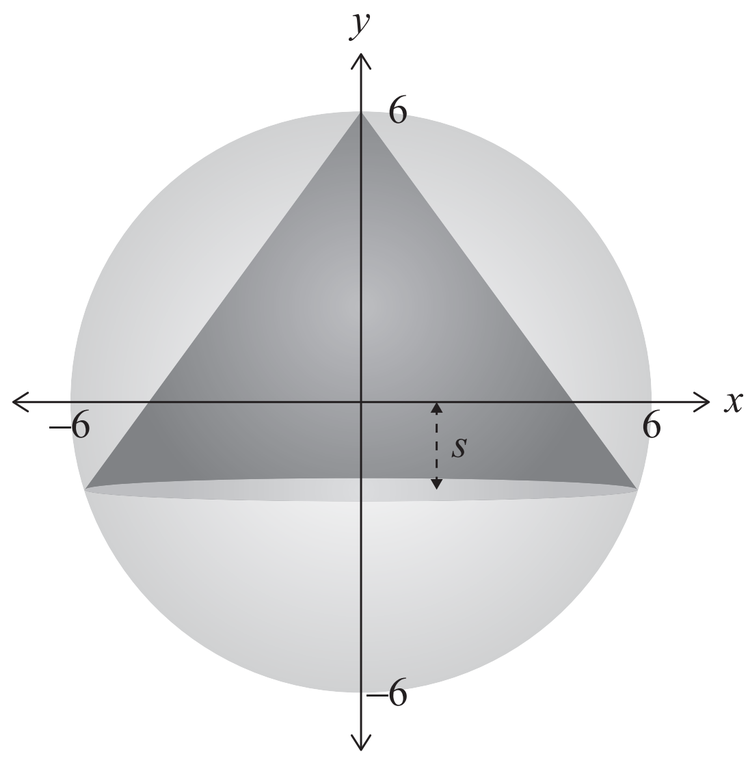
You must use calculus and show any derivatives that you need to find when solving this problem.

(e) A curve is defined by the function $f(x) = e^{-(x-k)^2}$.

Find, in terms of k, the x-coordinate(s) for which $f''(x) = 0$.

You must use calculus and show any derivatives that you need to find when solving this problem.

(e) A cone of height h and radius r is inscribed, as shown, inside a sphere of radius 6 cm.



The base of the cone is s cm below the x-axis.

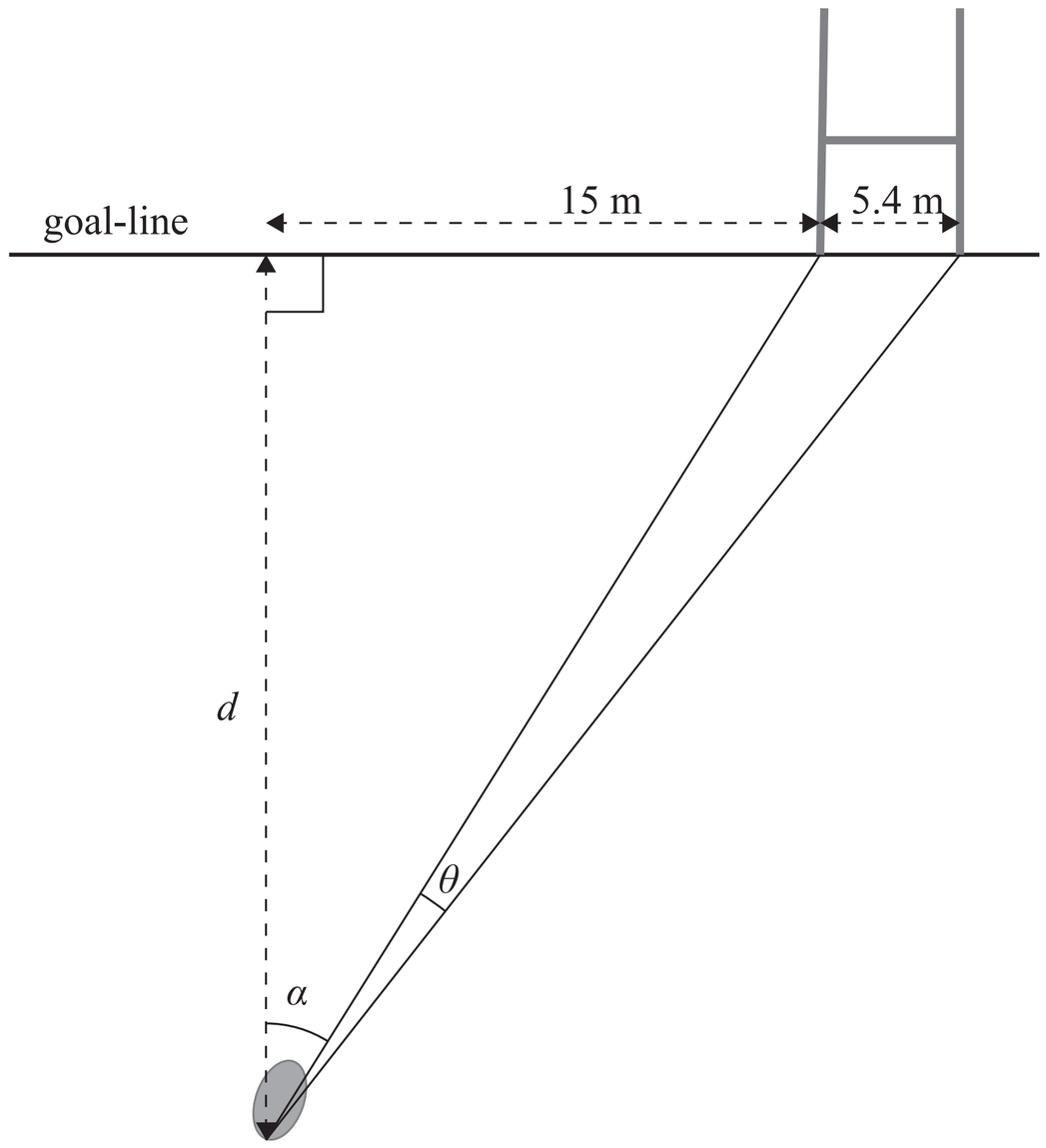
Find the value of s which maximises the volume of the cone.

You must use calculus and show any derivatives that you need to find when solving this problem.

You do not need to prove that the volume you have found is a maximum.

(e) In a rugby game, a try is scored 15 m from the left-hand goal-post. The conversion kick is taken at some point on the line perpendicular to the goal-line from the point where the try was scored, as shown in the diagram below.

The ball needs to pass between the goal-posts, which are 5.4 m apart.



Find the distance d from the goal-line that the conversion kick should be taken from in order to maximise the angle θ between the lines from the ball to the goal-posts.

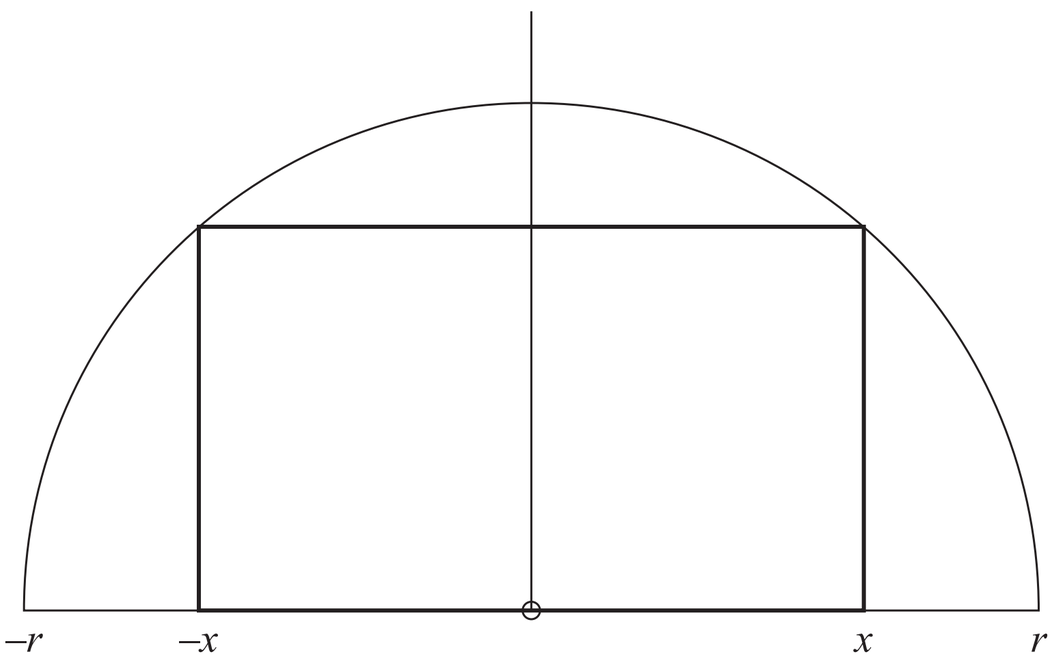
You must use calculus and show any derivatives that you need to find when solving this problem.

You do not need to prove that the angle you have found is a maximum.

(e) Find the values of a and b such that the curve $y = \dfrac{ax - b}{x^2 - 1}$ has a turning point at (3,1).

You must use calculus and show any derivatives that you need to find when solving this problem.

(e) A rectangle is inscribed in a semi-circle of radius r, as shown below.



Show that the maximum possible area of such a rectangle occurs when $x = \dfrac{r}{\sqrt{2}}$.

You do not need to prove that your solution gives the maximum area.

You must use calculus and show any derivatives that you need to find when solving this problem.

(e) For the function $y = e^x \cos kx$:

(i) Find $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$.

(ii) Find all the value(s) of k such that the function $y = e^x \cos kx$ satisfies the equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0 \text{ for all values of } x.$$

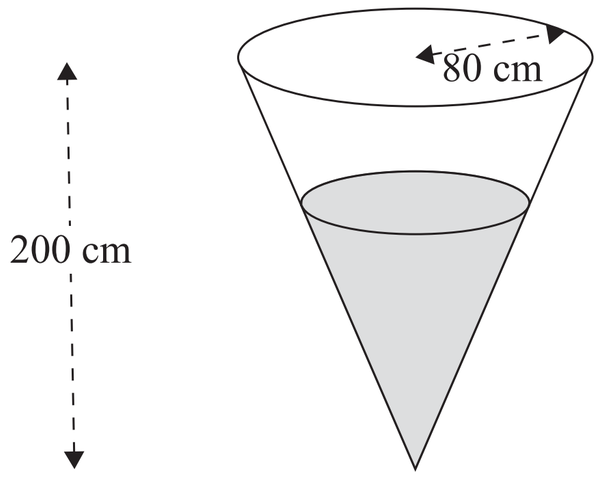
(e) A curve is defined by the parametric equations

$$x = t^3 + 1$$
$$y = t^2 + 1$$

Show that $\dfrac{\dfrac{d^2y}{dx^2}}{\left(\dfrac{dy}{dx}\right)^4}$ is a constant.

(e) A water tank is in the shape of an inverted right-circular cone.

The height of the cone is 200 cm and the radius of the cone is 80 cm.

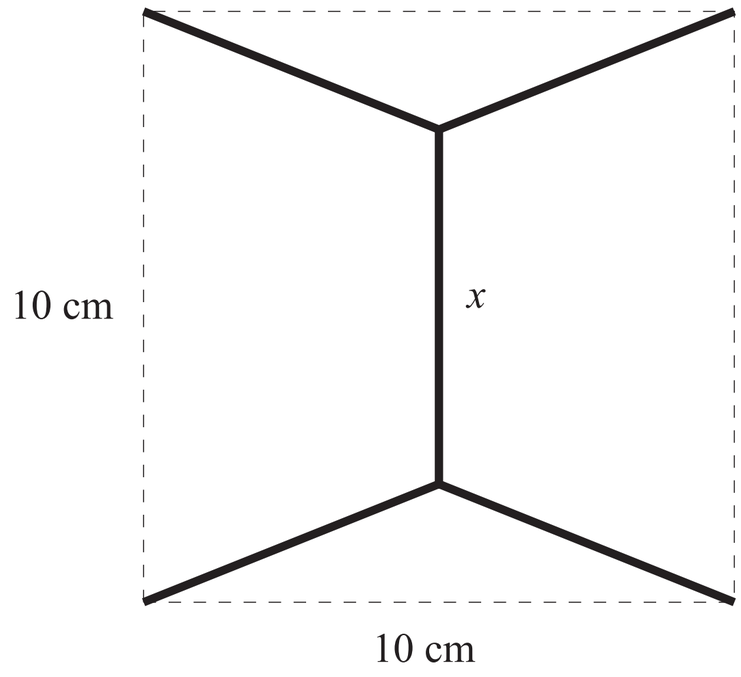


The tank is being filled with water at a rate of 150 cm^3 per second.

At what rate will the surface area of the water in the tank be increasing when the depth of water in the tank is 125 cm?

You must use calculus and show any derivatives that you need to find when solving this problem.

(e)



The above shape is made from wire. It has both vertical and horizontal lines of symmetry.

The ends of the shape are at the vertices of a square with a side length of 10 cm, as shown in the diagram above.

The length of the piece of wire through the centre of the shape is x cm.

Find the value(s) of x that enables the shape to be made with the minimum length of wire.

You do not need to prove that the length is a minimum.

You must use calculus and show any derivatives that you need to find when solving this problem.

(e) The volume of a sphere is increasing.

At the instant when the sphere's radius is 0.5 m, the surface area of the sphere is increasing at a rate of 0.4 $m^2\ s^{-1}$.

Find the rate at which the volume of the sphere is increasing at this instant.

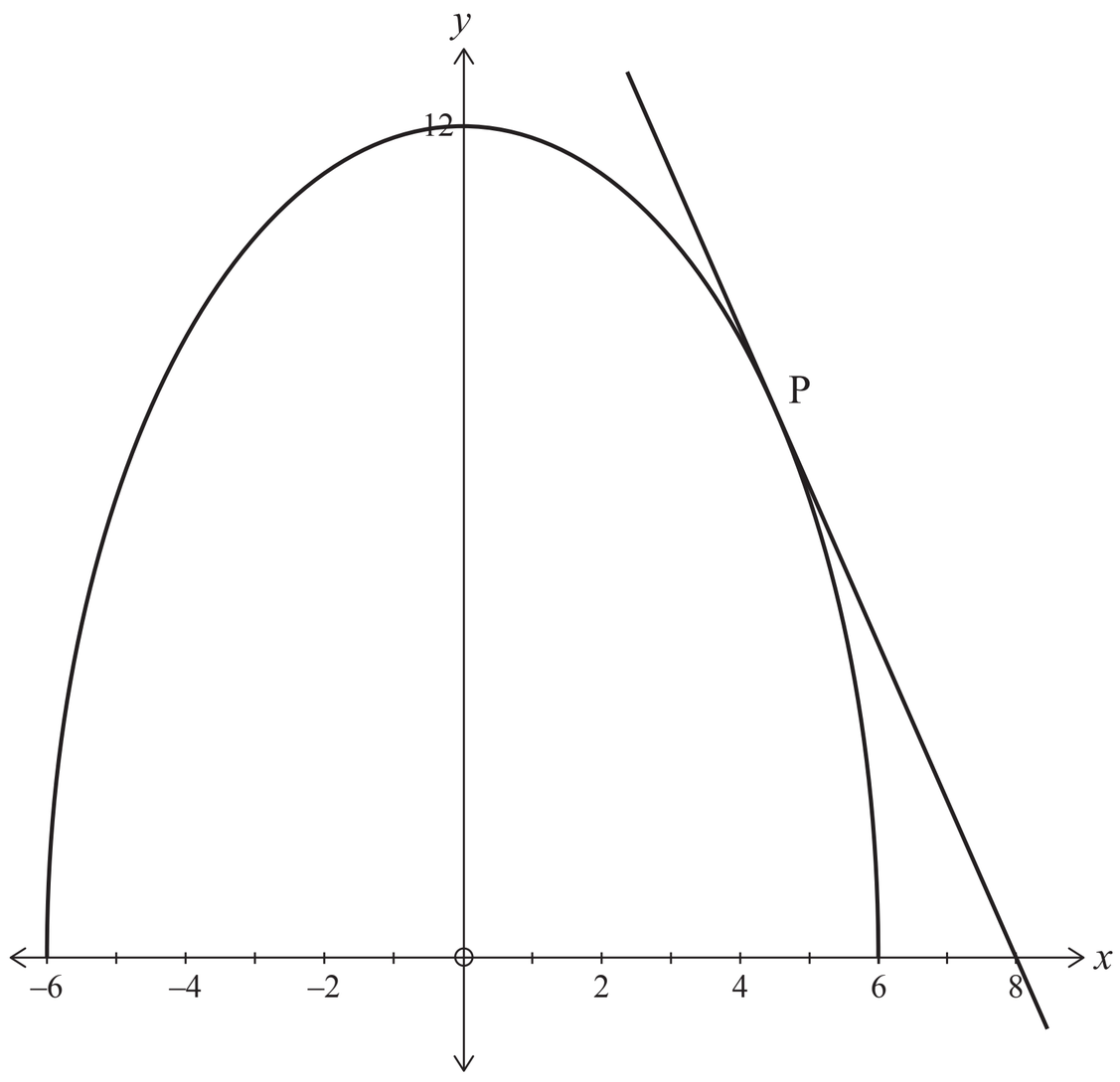
You must use calculus and show any derivatives that you need to find when solving this problem.

(e) If $y = e^u$ and $u = \sin 2x$ show that

$$\frac{d^2y}{dx^2} = \frac{d^2y}{du^2}\left(\frac{du}{dx}\right)^2 + \frac{dy}{du}\frac{d^2u}{dx^2}$$

You must use calculus and show any derivatives that you need to find when solving this problem.

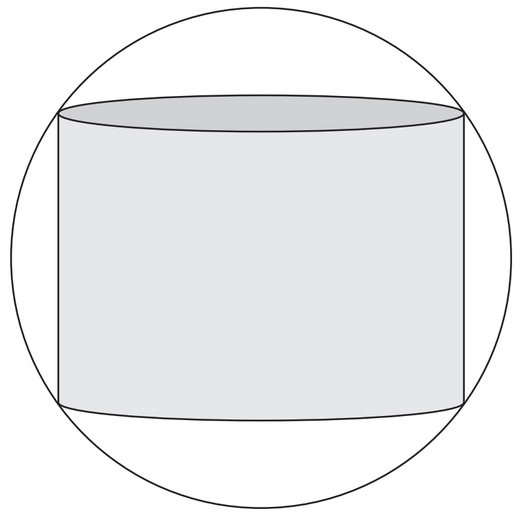
(e) The graph below shows the function $y = 2\sqrt{36 - x^2}$, and the tangent to that function at point P.
The tangent intersects the x-axis at the point (8,0).



Find the x-coordinate of point P.

You must use calculus and show any derivatives that you need to find when solving this problem.

(e) A cylinder of height h and radius r is inscribed, as shown to the right, inside a sphere of radius 20 cm.



Find the maximum possible volume of the cylinder.

You must use calculus and show any derivatives that you need to find when solving this problem.

You do not need to prove that the volume you have found is a maximum.

(e) A curve is defined by the parametric equations $x = \ln(t)$ and $y = 6t^3$ where $t > 0$.

The point P lies on the curve, and at point P, $\dfrac{d^2y}{dx^2} = 2$.

Find the exact coordinates of point P.

You must use calculus and show any derivatives that you need to find when solving this problem.

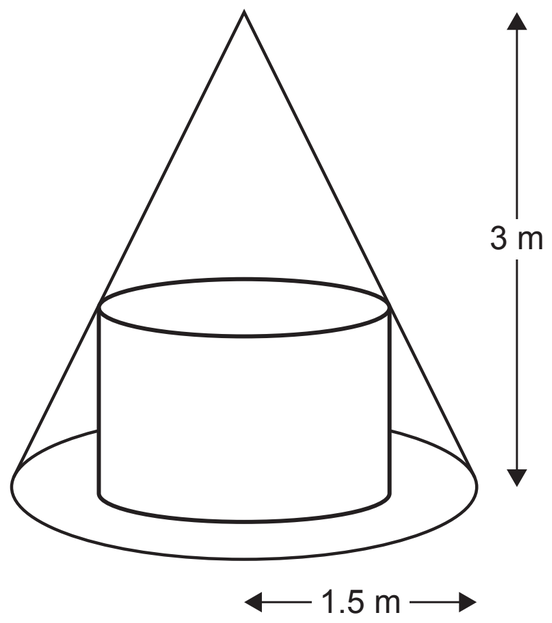
(e) A curve has the equation $y = \left(3x + 2\right)\mathrm{e}^{-2x}$.

Prove that $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\dfrac{\mathrm{d}y}{\mathrm{d}x} + 4y = 0$.

You must use calculus and show any derivatives that you need to find when solving this problem.

(e) A cone has a height of 3 m and a radius of 1.5 m.

A cylinder is inscribed in the cone, as shown in the diagram below.

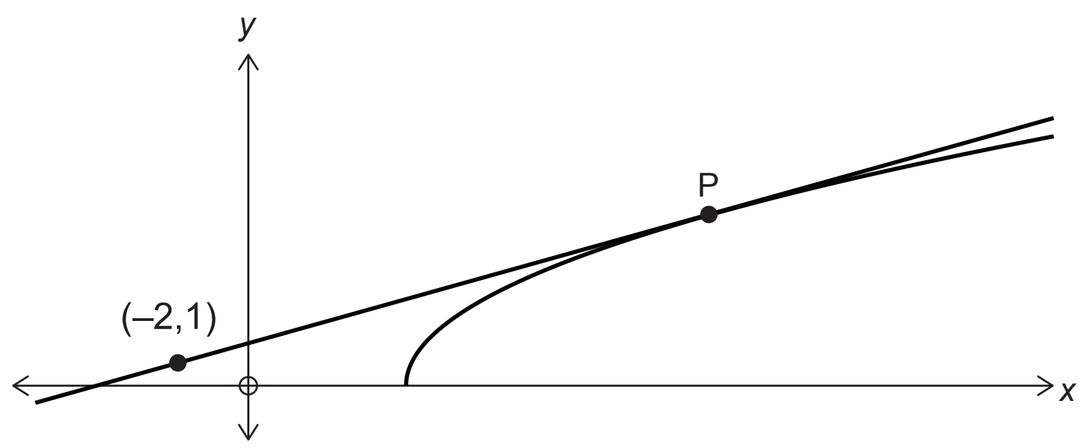


The base of the cylinder has the same centre as the base of the cone.

Prove that the maximum volume of the cylinder is π m^3.

You must use calculus and show any derivatives that you need to find when solving this problem.

(e) The graph below shows the curve $y = \sqrt{2x - 4}$, and the tangent to the curve at point P.
The tangent passes through the point (–2,1).

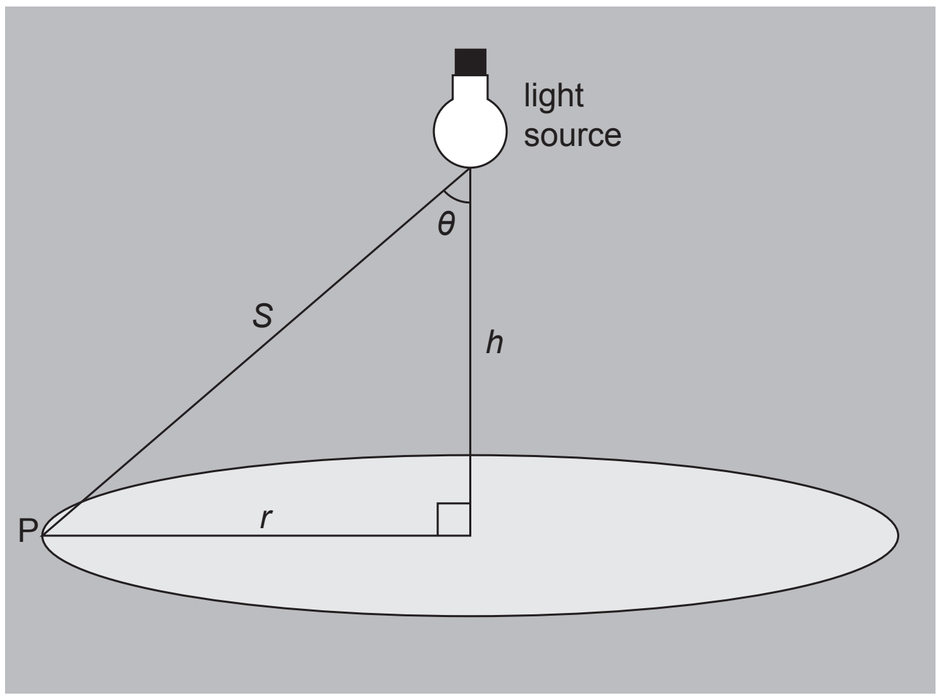


Find the coordinates of point P.

You must use calculus and show any derivatives that you need to find when solving this problem.

(e) A lamp is suspended above the centre of a round table of radius r.

The height, h, of the lamp above the table is adjustable.



Point P is on the edge of the table.

At point P the illumination I is directly proportional to the cosine of angle θ in the above diagram and inversely proportional to the square of the distance, S, to the lamp.

i.e. $I = \dfrac{k\cos\theta}{S^2}$, where k is a constant.

Prove that the edge of the table will have maximum illumination when $h = \dfrac{r}{\sqrt{2}}$.

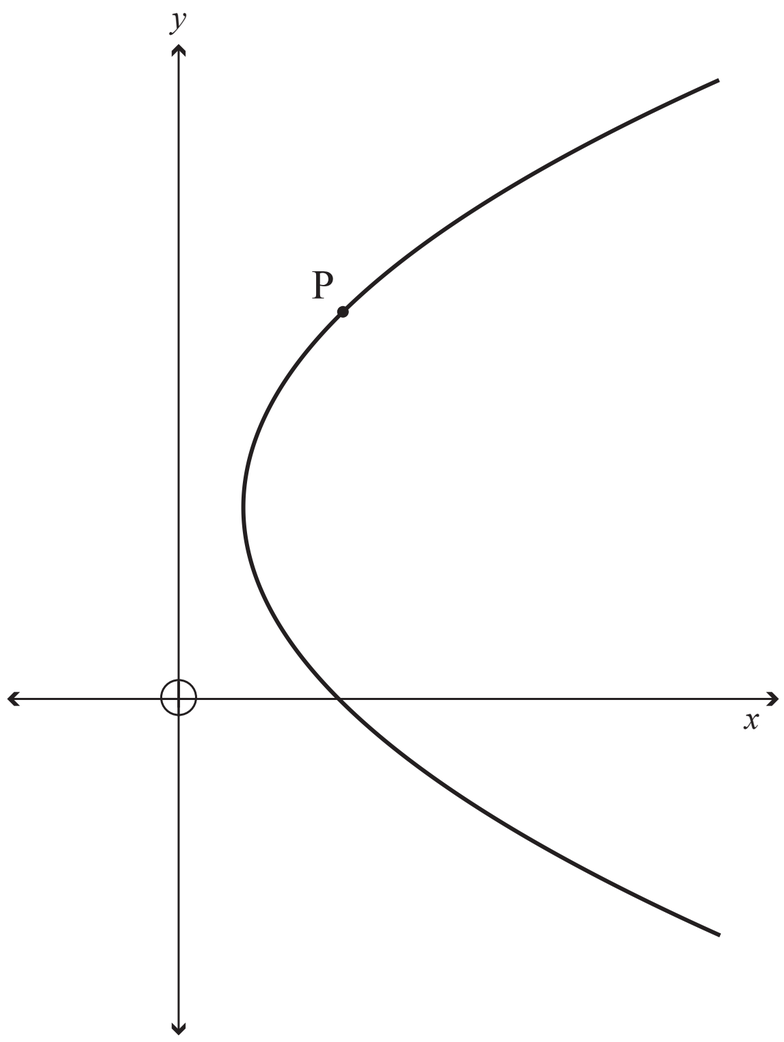
You do not need to prove that your solution gives the maximum value.

You must use calculus and show any derivatives that you need to find when solving this problem.

(e) If p is a positive real constant, prove that $y = e^{px^2}$ does not have any points of inflection.

You must use calculus and show any derivatives that you need to find when solving this problem.

(e) The curve with the equation $(y-5)^2 = 16(x-2)$ has a tangent of gradient 1 at point P.



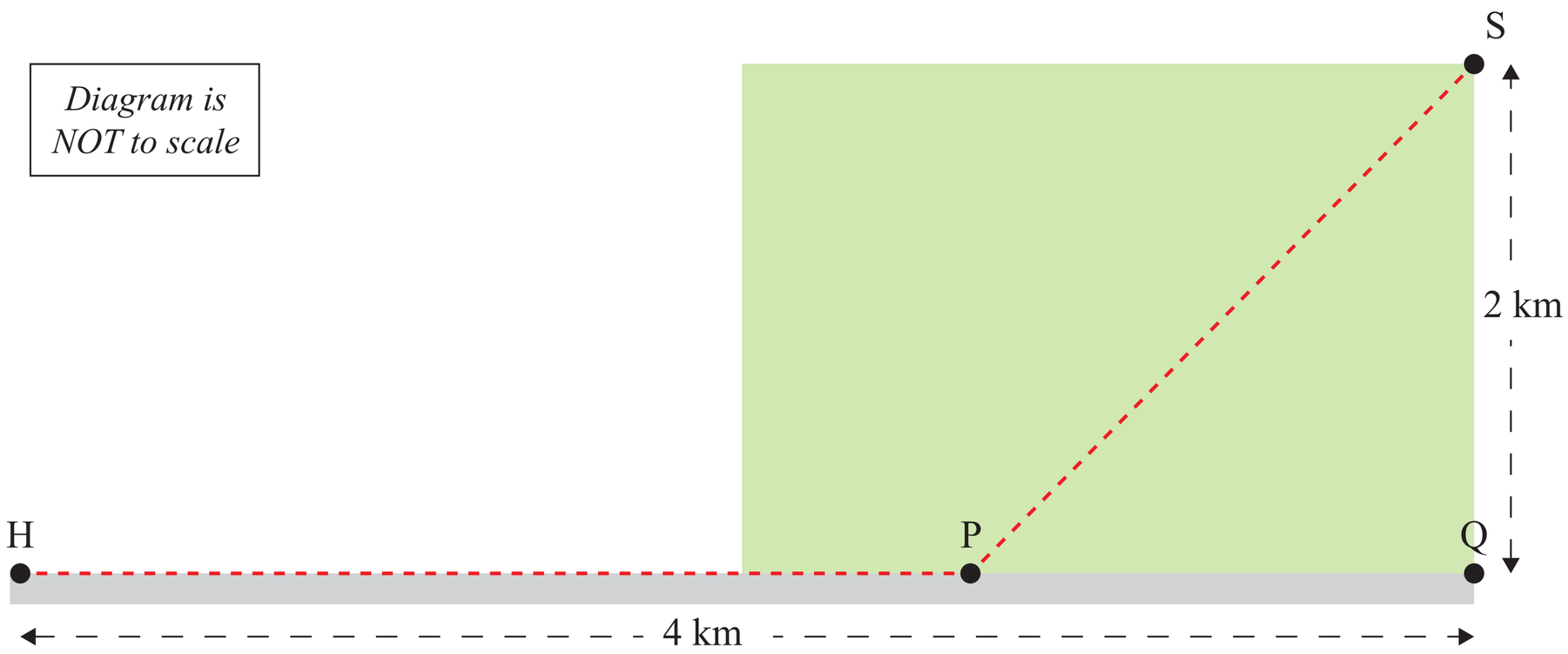
This tangent intersects the x and y axes at points R and S respectively.

Prove that the length RS is $7\sqrt{2}$.

You must use calculus and show any derivatives that you need to find when solving this problem.

(e) Megan cycles from her home, H, to school, S, each day.

Diagram is NOT to scale



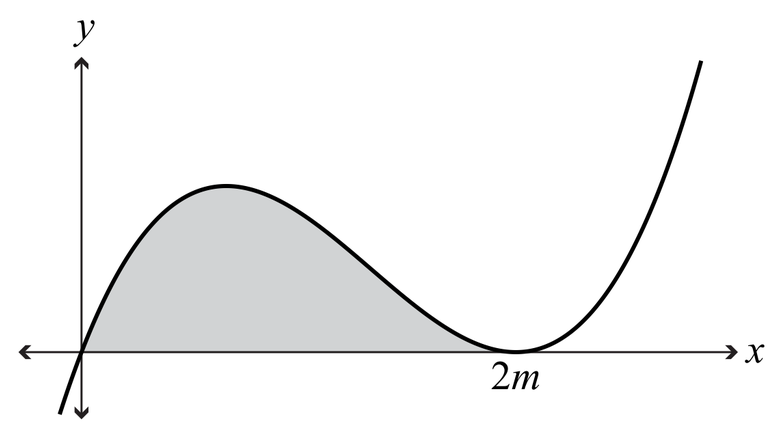
She rides along a path from her home to point P at a constant speed of 10 kilometres per hour.

At point P, Megan cuts across a park, heading directly to school. When cycling across the park, Megan can only cycle at 6 kilometres per hour.

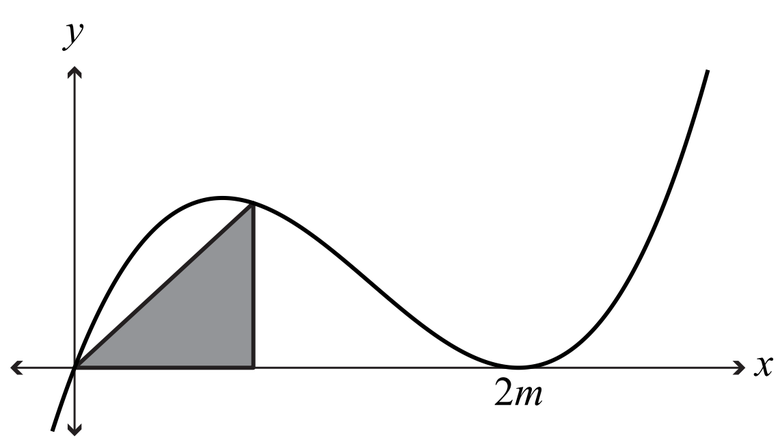
At what distance from her home should she choose to cut across the park in order to make her travelling time a minimum?

You must use calculus and show any derivatives that you need to find when solving this problem.

(e) The graph of $y = x(x - 2m)^2$, where $m > 0$, is shown.
The total shaded area between the curve and the x-axis from $x = 0$ to $x = 2m$ is given by $A = \dfrac{4m^4}{3}$.



A right-angled triangle is now constructed with one vertex at (0,0) and another on the curve $y = x(x - 2m)^2$, as shown below.



Show that the maximum area of such a triangle is $\dfrac{3}{8}$ of the total shaded area.

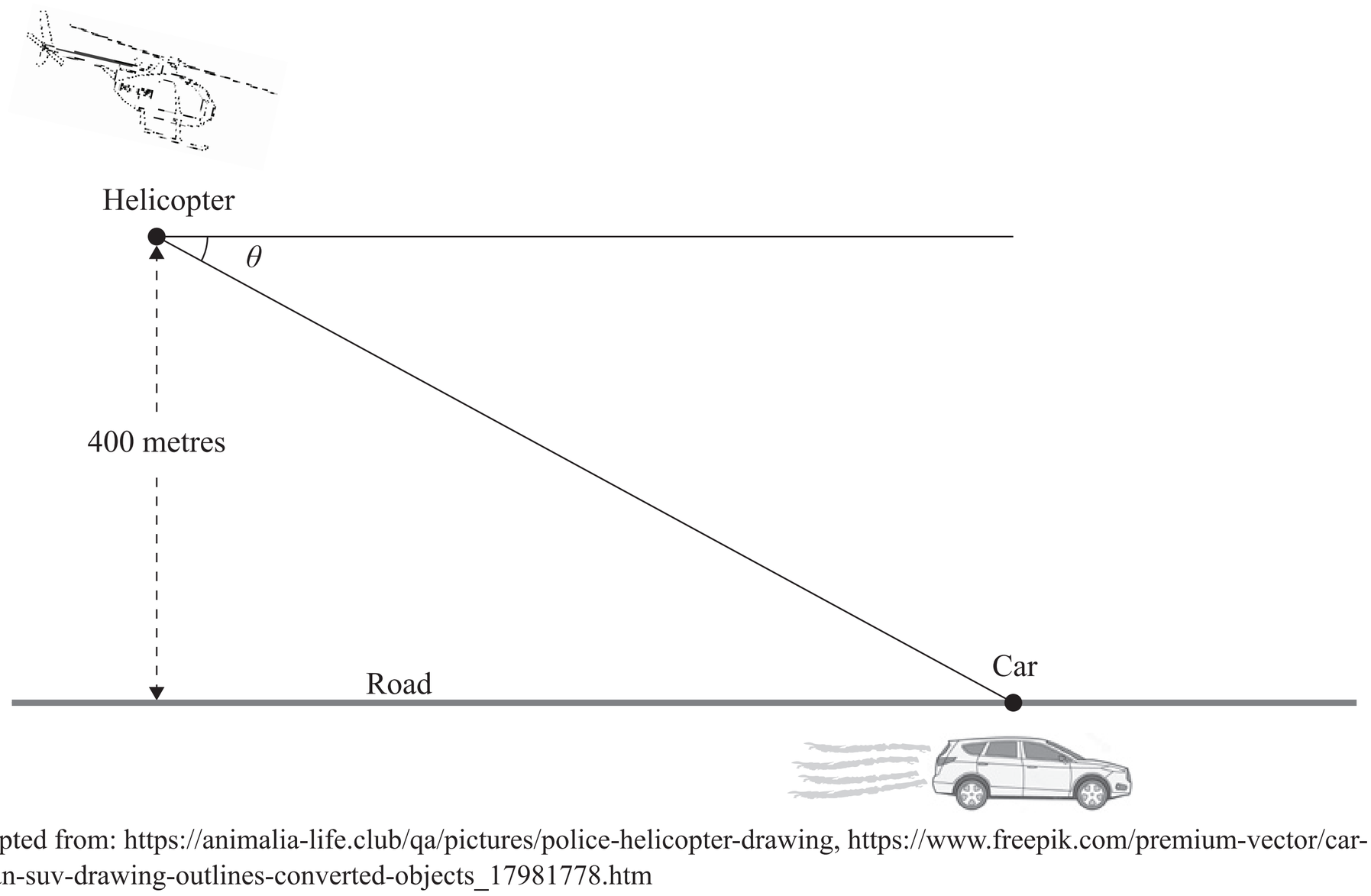
You must use calculus and show any derivatives that you need to find when solving this problem.

You do not have to prove that the area you have found is a maximum.

(e) A police helicopter is flying above a straight horizontal section of motorway chasing a speeding car.

The helicopter is flying at a constant speed of 72 m s^{-1} and at a constant height of 400 metres above the ground. The helicopter is attempting to catch up with the car.

When the direct distance from the helicopter to the car is 2500 metres, the angle of depression, θ between the horizontal and the line of sight from the helicopter to the car is increasing at a rate o 0.002 rad s^{-1}.



Adapted from: https://animalia-life.club/qa/pictures/police-helicopter-drawing, https://www.freepik.com/premium-vector/car-sedan-suv-drawing-outlines-converted-objects_17981778.htm

Calculate the speed of the **car** at this instant.

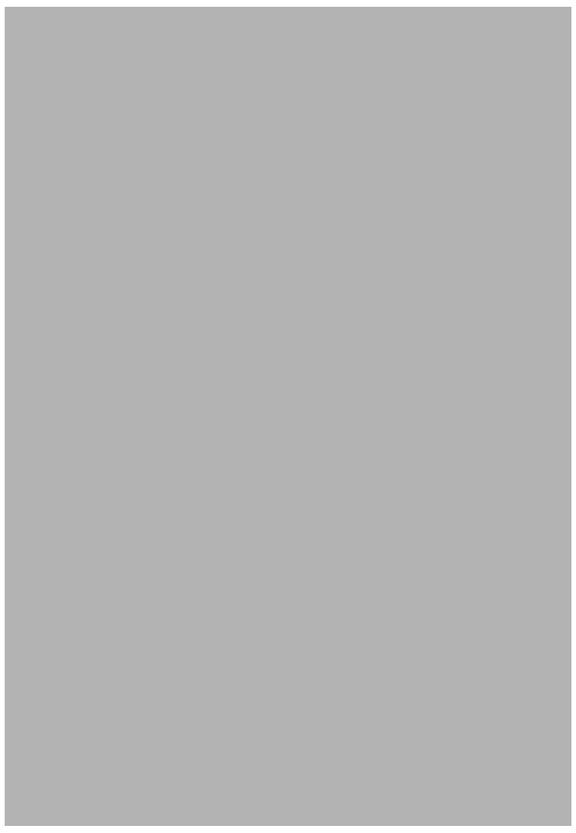
You must use calculus and show any derivatives that you need to find when solving this problem.

(e) A power line hangs between two poles.

The equation of the curve $y = f(x)$ that models the shape of the power line can be found by solving the differential equation:

$$a\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}$$

Use differentiation to verify that the function $y = \frac{a}{2}\left(\mathrm{e}^{\frac{x}{a}} + \mathrm{e}^{-\frac{x}{a}}\right)$ satisfies the above differential equation, where a is a positive constant.



Source: www.thelocalelectrician.com au/power-lines-power-pole-who-is-responsible/

(e) A curve is defined by the equation $y = \dfrac{2x^2 - 1 - 2x\ln x}{x}$, where $x > 0$.

The curve has a point of inflection at the point P.

Find the equation of the tangent to the curve at the point P.

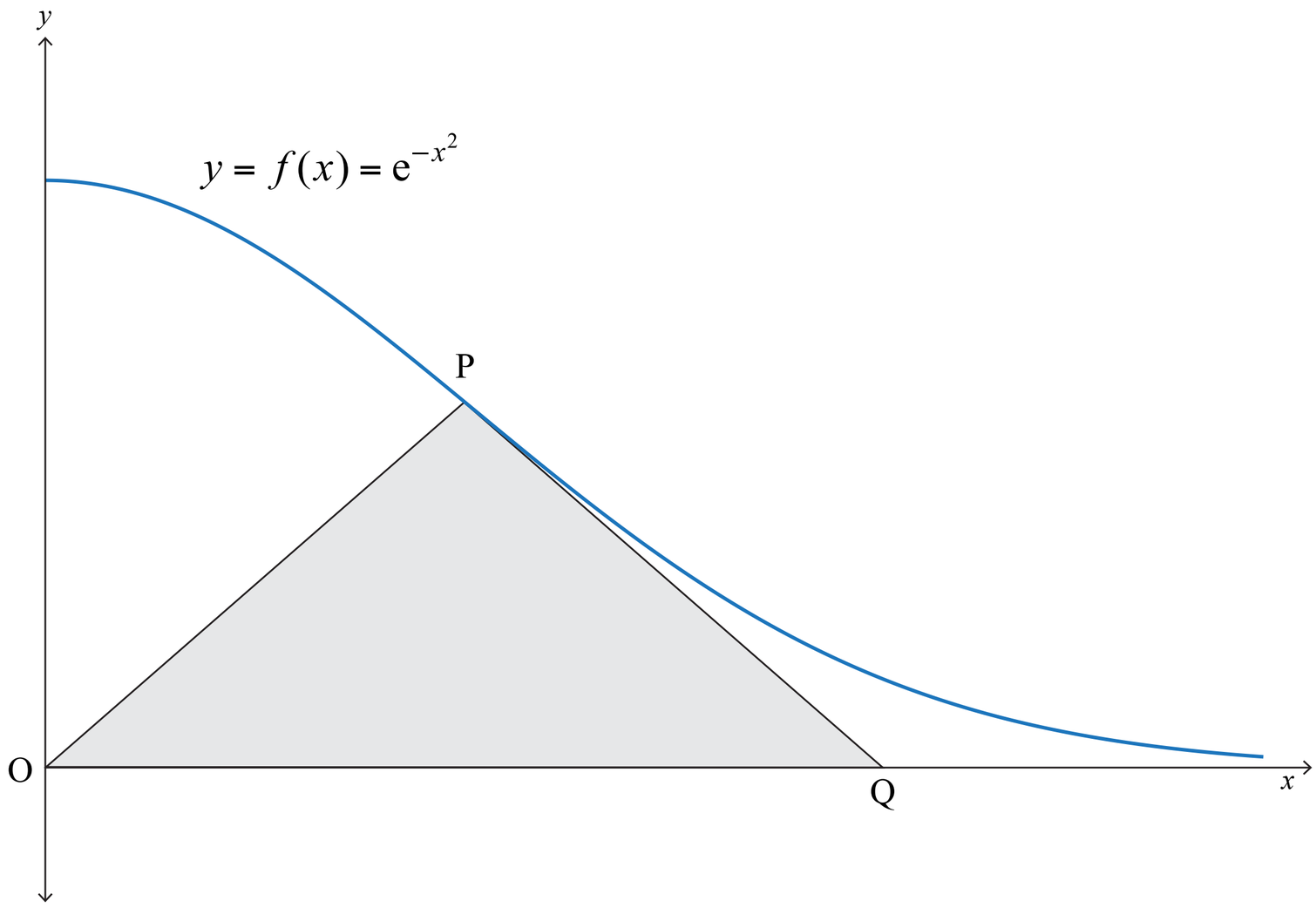
You must use calculus and show any derivatives that you need to find when solving this problem.

(e) The graph of the function $y = \dfrac{xe^{3x}}{2x+k}$, where k is a non-zero constant, **has a single turning poin** at **Q**.

Find the x-coordinate of the point Q.

You must use calculus and show any derivatives that you need to find when solving this problem.

(e) The diagram below shows part of the graph of the function $f(x) = e^{-x^2}$, where $x \geq 0$.



The point P lies on the curve and the point Q lies on the x-axis so that OP = PQ, where O is the origin.

Prove that the largest possible area of the triangle OPQ is $\frac{1}{\sqrt{2e}}$.

You do not need to show that the area you have found is a maximum.

You must use calculus and show any derivatives that you need to find when solving this problem.

(e) A curve is defined by the pair of parametric equations:

$x = 3t^2 + 6t + 1$ and $y = 2t^3$

Find the coordinates of any points of inflection on this curve that are also stationary points.

You may assume that any inflection point(s) found are actually inflection points.

You must use calculus and show any derivatives that you need to find when solving this problem.

(e) The radius of curvature of a function is a measure of how much the curve is bending at a given point.

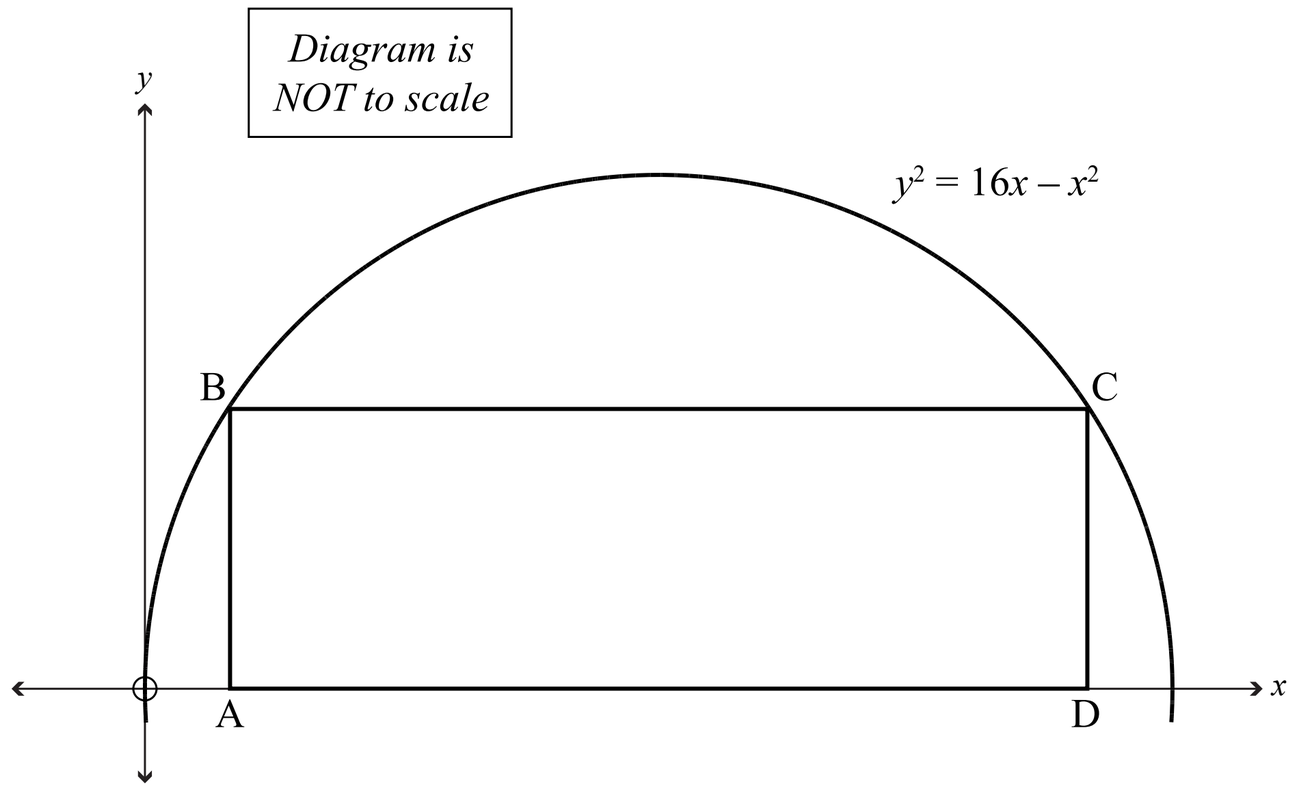
The formula to calculate the radius of curvature is given by $p = \dfrac{\left(1 + \left(\dfrac{dy}{dx}\right)^2\right)^{\frac{3}{2}}}{\dfrac{d^2y}{dx^2}}$

Find the radius of curvature for the curve $y = \cos^2(2x)$ when $x = \dfrac{\pi}{3}$.

You must use calculus and show any derivatives that you need to find when solving this problem.

(e) The diagram below shows part of the symmetrical graph $y^2 = 16x - x^2$.

A rectangle ABCD is drawn inside the curve with its vertices B and C lying on the curve.



Find the **length AD** so that the rectangle has its maximum area.

You can assume that the value found is a maximum.

You must use calculus and show any derivatives that you need to find when solving this problem.